LESSON 7.6

Determine Distance on the Coordinate Plane

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the distance between two points on the coordinate plane using absolute value.

WHY THIS MATTERS Captain Vega has two treasure locations on her map: a buried chest at (-3, 2) and a hidden cave at (4, 2). Both are at the same height (y = 2) on the map. She needs to figure out how far apart they are so she knows how much rope to bring. Another pair of treasures is at (1, -3) and (1, 5) — same column, different rows. How far apart is each pair?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Distance *Spanish: Distancia*	How far apart two points are. It is never negative.	always a positive length
Absolute value *Spanish: Valor absoluto*	How far a number is from zero. It is never negative.	$\|-7\| = 7$
Horizontal distance *Spanish: Distancia horizontal*	How far apart two points are going left or right.	uses the x-coordinates
Vertical distance *Spanish: Distancia vertical*	How far apart two points are going up or down.	uses the y-coordinates
Coordinate plane *Spanish: Plano cartesiano*	A grid with a line going across and a line going up to plot points.	x-axis across, y-axis up
Integer *Spanish: Número entero*	Whole numbers and their opposites, like -2, -1, 0, 1, 2.	..., -3, -2, -1, 0, 1, 2, 3, ...

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Each unit is 1 meter. Find the distance from P(−9, 4) to Q(3, 4), and from P to R(−9, 11).

1. P and Q share y = 4, so use x. −9 and 3 have opposite signs: 9 + 3 = 12 m.
2. P and R share x = −9, so use y. 4 and 11 are both positive: 11 − 4 = 7 m.

Answer: PQ = 12 m and PR = 7 m

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Now let's find the distance from (2, 3) to (2, 8). What do these points share? The same x = 2, so it is a vertical line.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The two points share the _____ -coordinate, so the segment runs _____ .
- I used absolute value because _____ .
- The points sit on _____ sides of the axis, so I _____ the two distances.
- The distance is _____ units, which means _____ on this map.

Word bank distance | absolute value | horizontal | vertical | coordinate plane | axis | units | add | subtract | x-coordinate | y-coordinate | segment

WATCH OUT!

A common mistake

A common mistake in Distance on the Coordinate Plane is subtracting the coordinates as if both points were on the same side of zero, instead of adding their absolute values when the points are on opposite sides. For example, to find the distance between (-6, 3) and (2, 3), a student might subtract 6 − 2 = 4, but since -6 and 2 are on opposite sides of zero, the absolute values must be added instead: |-6| + |2| = 6 + 2 = 8. The correct distance is 8 units, not 4. Before you submit, ask: 'Are these two points on opposite sides of zero

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Level 0 · Most support · every hint printed

MASTERY CHECK

- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Both points in each row share a coordinate. Find the distance.

Point 1	Point 2	Shared Coordinate	Distance
(-5, 1)	(3, 1)	y = 1	
(2, -4)	(2, 6)	x = 2	
(-3, -2)	(-3, 4)	x = -3	
(-1, 5)	(4, 5)	y = 5	

STEPS TO TRY

1. Each row shares a coordinate, so every distance is a straight horizontal or vertical count.
2. Subtract the coordinates that DIFFER and take the absolute value. If the signs differ, add the two absolute values.
3. Row 1: the y's match, so count from x = -5 to x = 3. Go -5 to 0, then 0 to 3, and add the two counts.

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

What is the distance between (-4, 3) and (2, 3)?

(A) 2 units
(B) 4 units
(C) 6 units
(D) 8 units

STEPS TO TRY

1. Both points have y = 3, so they sit on the same horizontal line — only the x-values differ.
2. Find the gap between x = -4 and x = 2. The path crosses zero, so add the two distances |-4| and |2|.
3. Count from -4 to 0 (that's 4 units), then from 0 to 2. Add the two parts together.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

What is the distance between (5, -2) and (5, 4)?

(A) 2 units
(B) 4 units
(C) 6 units
(D) 10 units

STEPS TO TRY

1. Both points share x = 5 — the segment is VERTICAL, so compare the y-values -2 and 4.
2. The path crosses zero: split it into a below-zero part and an above-zero part using absolute values.
3. From y = -2 up to 0 is 2 units; from 0 up to 4 is 4 more. Add the parts.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A rectangle has vertices at (1, 1), (5, 1), (5, 4), (1, 4). What is its perimeter?

(A) 8 units
(B) 12 units
(C) 14 units
(D) 20 units

STEPS TO TRY

1. Perimeter is the distance all the way around — you need the width AND the height first.
2. Width: subtract the x-values 5 and 1. Height: subtract the y-values 4 and 1. Then add all four sides.
3. Find 5 − 1 for the width and 4 − 1 for the height. Now double each and add — that covers all four sides.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

EXPLAIN YOUR THINKING

How did you use coordinates and absolute value to find the distance between two points in the same row or column?

SENTENCE STARTER The points ___ and ___ share the same ___ -coordinate. To find the distance, I found |___| + |___| = ___, which equals ___ units.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help